Randomized Algorithm

- Deterministic algorithm is an algorithm in which for given particular input it will always produce the same output, with the underlying machine always passing through the same sequence of states.

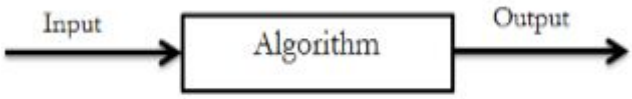


Figure 1

- To overcome the computation problem of exploitation of running time of a deterministic algorithm, randomized algorithm is used.
- Randomized algorithm uses uniform random bits also called as pseudo random number as an input to guides its behavior (Output).
- Randomized algorithms rely on the statistical properties of random numbers.
- One example of a randomized algorithm is quicksort.
- It tries to achieve good performance in the average case.
- Randomized algorithm is also called as probabilistic algorithm.

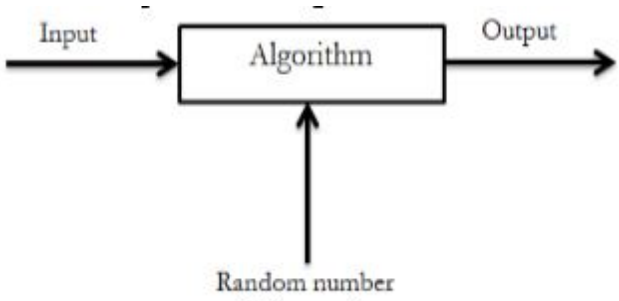


Figure 2

- Randomized algorithms are algorithms that make use of use of

random decisions.

- We call an algorithm randomized if its behavior is determined not only by its input but also by values produced by a random-number generator.
- Most programming environments offer a (deterministic) pseudorandom-number generator: it returns numbers that ”look” statistically random.
- We typically refer to the analysis of randomized algorithms by talking about the expected cost (ex: the expected running time).
- We can use probabilistic analysis to analyze randomized algorithms.

Two Types of Randomized Algorithms

Randomized algorithms can be classified into two types:

- **Las Vegas Algorithm**: Always outputs the correct answer but running time is random.
- Analysis: Compute expected running time.
- **Monte Carlo Algorithm:** Always run in a fixed amount of time but result may be incorrect.

Advantage of Randomized Algorithm:

1. The algorithm is usually simple and easy to implement.
2. The algorithm is fast with very high probability, and/or it produces optimum output with very high probability.